Review chapter 12

Derivative for Multivariable Functions

12.1 Functions of Two or More Variables

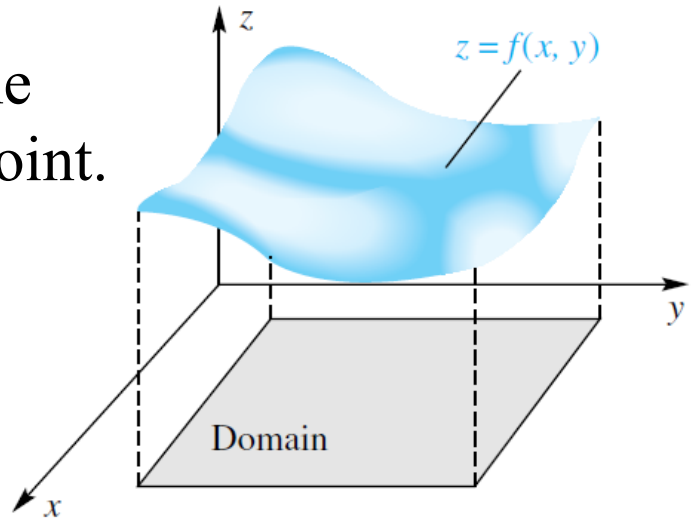
Graphs of Functions of Two Variables

The **graph** of a function f of two variables with domain D is the set of all points (x, y, z) in R^3 such that $z = f(x,y)$ and (x,y) is in D.

$$\{(x, y, z) \in R^3 : z = f(x, y), (x, y) \in D\}.$$

The graph is normally a surface S in three space.

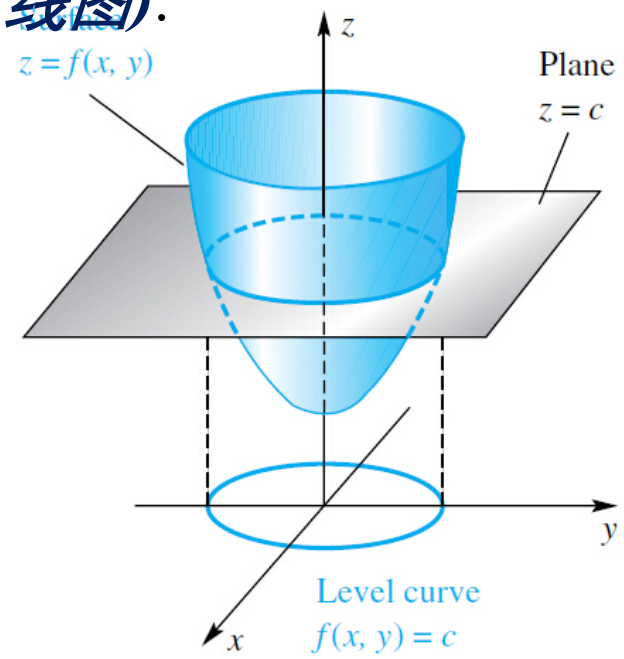
Each line perpendicular to the xy-plane intersects the surface in at most one point.

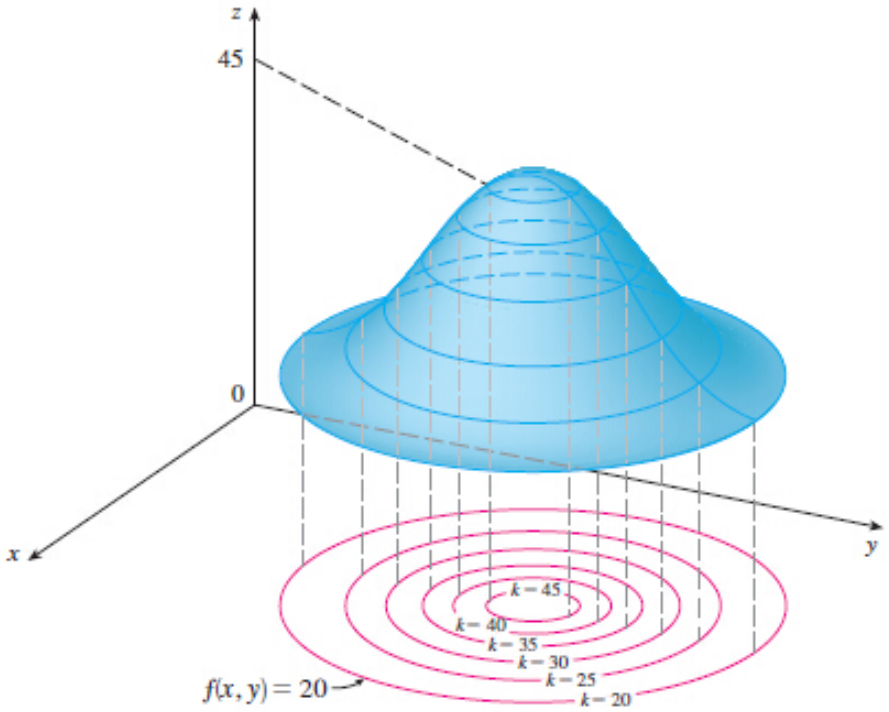


Level Curves

The **level curves** (等值线)of a function f of two variables are the curves in xy-plane with equation $f(x, y) = c$, where c is a constant (in the range of f).

Geometrically, each horizontal plane $z = c$ intersects the surface in a curve. The projection of this curve on xy-plane is a level curve. A collection of such curves is called a ***contour map(等高线图)***.





Functions of Three or More Variables

Definition A function of three variables is a function f that assigns to each ordered triple (x, y, z) in a domain $D \subset R^3$ a unique real number denoted by $f(x, y, z)$.

Domain: The set $D \subset R^3 = \{(x, y, z) : x, y, z \in R\}$.

Range: The set of values that f takes on, that is

$$\{f(x, y, z) : (x, y, z) \in D\} \subset R.$$

Level Surfaces

The **level surfaces** of a function f of three variables are the surfaces in three space with equations $f(x, y, z) = c$, where c is a constant (in the range of f).

12.2 Partial Derivatives

Calculation of Partial Derivatives

Rules for Finding Partial Derivatives of $z = f(x, y)$.

1. To find f_x, regard y as a constant and differentiate $f(x, y)$ with respect to x.
2. To find f_y, regard x as a constant and differentiate $f(x, y)$ with respect to y.

Notations for Partial Derivatives

If $z=f(x, y)$, we use the following alternative notations:

$$f_x(x\ ,y\ ) = \frac{\partial z}{\partial x} = \frac{\partial}{\partial x} f(x,y),\ \ f_y(x\ ,y\ ) = \frac{\partial z}{\partial y} = \frac{\partial}{\partial y} f(x,y)$$

$$f_x(x_0, y_0) = \left.\frac{\partial z}{\partial x}\right|_{(x_0, y_0)}, \qquad f_y(x_0, y_0) = \left.\frac{\partial z}{\partial y}\right|_{(x_0, y_0)}$$

∂ is called the partial derivative sign.

$\dfrac{\partial}{\partial x}$ and $\dfrac{\partial}{\partial y}$ represent linear operators,

much like the linear operators D_x or $\dfrac{d}{dx}$.

Formulas of Derivatives

$$(C)' = 0 \qquad (x^{\mu})' = \mu x^{\mu-1}$$

$$(\mathrm{e}^{x})' = \mathrm{e}^{x} \qquad (a^{x})' = a^{x}\ln a$$

$$(\ln x)' = \frac{1}{x} \qquad (\log_a x)' = \frac{1}{x\ln a}$$

$$(\sin x)' = \cos x \qquad (\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x \qquad (\cot x)' = -\csc^2 x$$

$$(\sec x)' = \sec x \cdot \tan x \qquad (\csc x)' = -\csc x \cdot \cot x$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}} \qquad (\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2} \qquad (\operatorname{arc}\cot x)' = -\frac{1}{1+x^2}$$

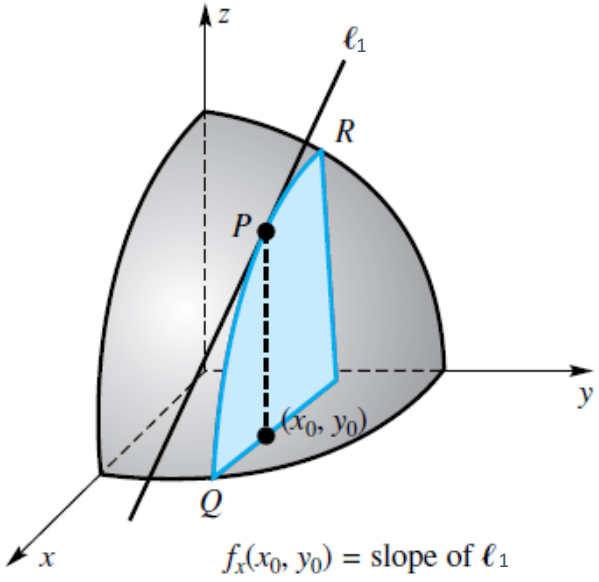
Geometric Interpretation of Partial Derivatives

The equation of $z=f(x, y)$ represents a surface S (the graph of f).

The point $P(x_0, y_0, f(x_0, y_0))$ lies on the surface.

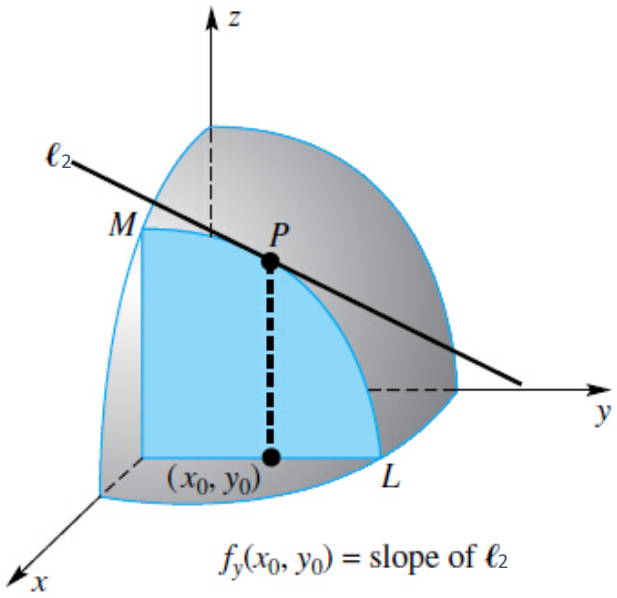
The plane $y = y_0$ intersects this surface in the plane curve QPR.

Then $f_x(x_0, y_0)$ can be interpreted geometrically as the slope of the tangent line l_1 to this curve at P.



Geometric Interpretation of Partial Derivatives

Similarly, the plane $x = x_0$ intersects this surface in the plane curve *LPM*. Then $f_y(x_0, y_0)$ can be interpreted as the slope of the tangent line l_2 to this curve at *P*.



Higher Partial Derivatives

Second partial derivatives of $f(x, y)$.

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} \qquad f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}$$

$$f_{xy} = (f_x)_y = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} \qquad f_{yx} = (f_y)_x = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

Derivatives for Functions of More Than Two Variables

Example Let $f(x,y,z) = xy^2z^3$. Find $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$.

Solution

$$f_x(x,y,z) = y^2z^3$$

$$f_y(x,y,z) = 2xyz^3$$

$$f_z(x,y,z) = 3xy^2z^2$$

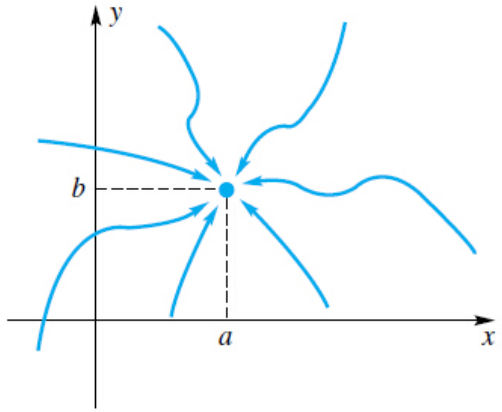
➢12.3 Limits and Continuity

- **Limit of a Function of Two Variables**

If the values of $f(x,y)$ get closer and closer to L as the point (x,y) approaches (a,b) along any path in the domain of f, then we say that the limit of $f(x,y)$ as (x,y) approaches (a,b) is L and we write $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L.$

In other words,

We can make the values of $f(x,y)$ as close to L as we like by taking (x,y) sufficiently close to the point (a,b) but not equal to (a,b).



Example 1 Evaluate the following limits if they exist:

$$(a)\ \lim_{(x,y)\to(1,2)}(x^2y+3y), \qquad (b)\ \lim_{(x,y)\to(0,0)}\frac{x^2+y^2+1}{x^2-y^2}.$$

Solution $(a)\ \lim_{(x,y)\to(1,2)}(x^2y+3y)=1^2\cdot 2+3\cdot 2=8.$

(b) Since $\lim_{(x,y)\to(0,0)}(x^2-y^2)=0$ and $\lim_{(x,y)\to(0,0)}(x^2+y^2+1)=1,$

the given limit does not exist.

Example2 Evaluate the following limits if they exist:

$$(a)\ \lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{3x^2+3y^2}, \qquad (b)\ \lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2}.$$

Solution (a) Let $u = x^2 + y^2$. Then $(x, y) \to (0,0) \Leftrightarrow u \to 0$, thus

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{3x^2+3y^2} = \lim_{u\to 0} \frac{\sin u}{3u} = \frac{1}{3}.$$

(b) Since $0 < \left|\dfrac{3x^2y}{x^2+y^2}\right| = 3\dfrac{x^2}{x^2+y^2}|y| < 3|y| \to 0$ as $(x, y) \to (0,0)$,

by squeeze theorem, we conclude that

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0.$$

- **Two Paths Test** **to judge discontinuity**

If $f(x,y) \to L_1$ as $(x,y) \to (a,b)$ along a path C_1 and $f(x,y) \to L_2$ as $(x,y) \to (a,b)$ along a path C_2 , where $L_1 \neq L_2$, then $\lim\limits_{(x,y)\to(a,b)} f(x,y)$ does not exist.

If we can find two different paths of approach along which the function $f(x,y)$ has different limits, then the limit $\lim\limits_{(x,y)\to(x_0,y_0)} f(x,y)$ does not exist.

Example 3 Show that the function f defined by

$$f(x,y)=\frac{x^2-y^2}{x^2+y^2}$$

has no limit at the origin.

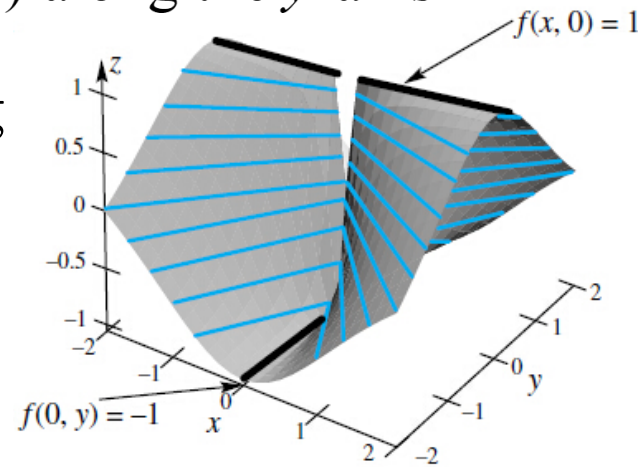
Solution Along the x-axis, $y = 0$, then $f(x, 0)=1$. Thus

$$f(x,y)\to 1 \quad \text{as } (x,y)\to(0,0) \text{ along the } x\text{-axis}$$

Along the y-axis, $x = 0$, then $f(0, y)= -1$. So

$$f(x,y)\to -1 \quad \text{as } (x,y)\to(0,0) \text{ along the } y\text{-axis}$$

Since f has two different limits along two different lines, the given limit does not exist.



- **Continuity**

1. Intuitively, continuity means that the graph of f has no hole or break.

2. **Polynomial functions** (多项式函数) of two variables are continuous everywhere, since they are sums and products of the continuous functions x, y, and c.

   For example $f(x,y) = x^2y^3 + x^3y^2 + 3x + 4y - 6$

   is continuous everywhere in the xy-plane.

3. **Rational functions**(有理函数) of two variables are quotients of polynomial functions and thus are continuous wherever the denominator is not zero.

   For example $f(x,y) = \dfrac{2x+3y}{x^2+y^2}$ is continuous except

   at (0, 0).

➢12.4 Differentiability

1. Tangent Plane of the surface

The equation of the tangent plane to $z = f(x, y)$ ***at*** $P(x_0, y_0, f(x_0, y_0))$ *is*

$$\boxed{\begin{aligned} & f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - f(x_0, y_0)) = 0 \\ \Leftrightarrow\ & z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \end{aligned}}$$

Sufficient Condition for Differentiability

Theorem A If the partial derivatives $f_x(x, y)$ and $f_y(x, y)$ of $f(x, y)$ are all continuous functions, then $f(x, y)$ is a differentiable function.

Theorem C If $f(x, y)$ is differentiable at (x_0, y_0), then $f(x, y)$ is continuous at (x_0, y_0).

- **Gradient of Functions of Two Variables**

Definition For $z = f(x, y) = f(\mathbf{p})$, the vector

$$\left\langle f_x(\mathbf{p}), f_y(\mathbf{p}) \right\rangle = f_x(\mathbf{p})\mathbf{i} + f_y(\mathbf{p})\mathbf{j}, \text{ or,}$$

$$\left\langle f_x(x, y), f_y(x, y) \right\rangle = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

is called the ***gradient*** of f at $\mathbf{p}$ and is denoted by $\nabla f(\mathbf{p}) = \nabla f(x, y)$.

"∇" is read "del" , and is called "del operator".

➢12.5 Directional Derivatives(方向导数) and Gradients(梯度)

1. **Gradient and Directional Derivatives**

2. **Gradient and Maximum rate of Change (Geometric Meaning of the Gradient)**

4. **Gradient and Level curve**

2. Computation of directional derivatives

Theorem A

Let $f(x, y)$ be differentiable at $\mathbf{P}(x, y)$. Then $f(x, y)$ has a directional derivative at $\mathbf{P}$ in the direction of the unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$ and

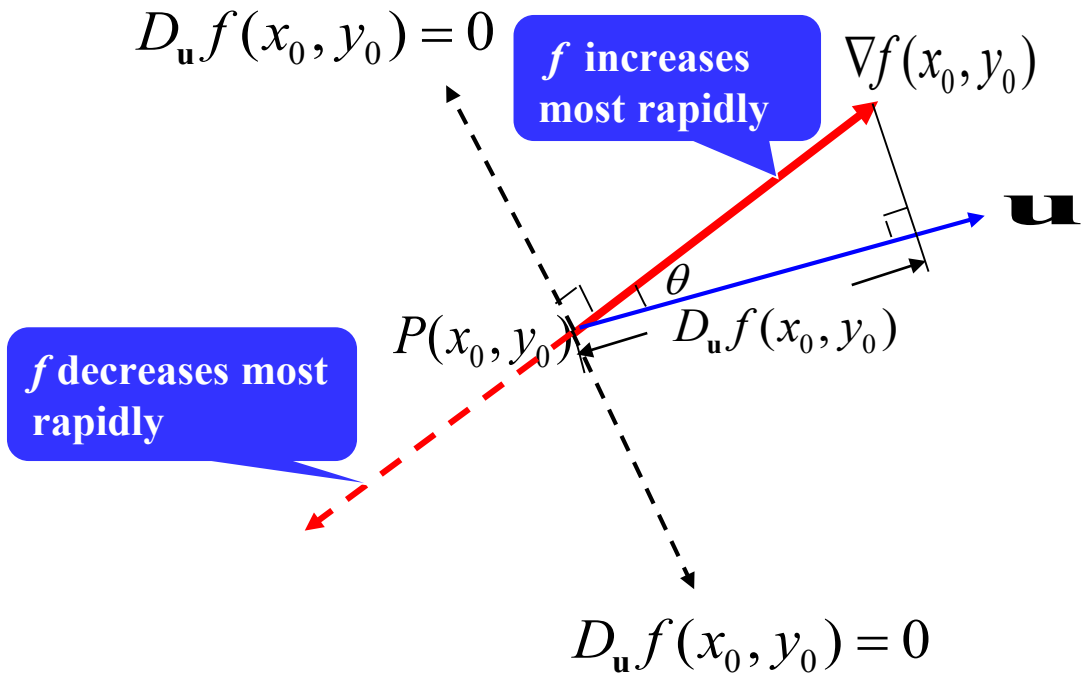
$$D_u f(x, y) = f_x(x, y) \cdot u_1 + f_y(x, y) \cdot u_2 = \nabla f \cdot \mathbf{u}$$

Directional derivatives of functions of three variables $f(x, y, z)$

If f is differentiable, just as with functions of two variables,

$$\boxed{\begin{aligned} D_{\mathbf{u}} f(\mathbf{p}) &= \nabla f(\mathbf{p}) \cdot \mathbf{u} \\ &= f_x(x, y, z)u_1 + f_y(x, y, z)u_2 + f_z(x, y, z)u_3. \end{aligned}}$$

Theorem B A function increases most rapidly at **p** in the direction of the gradient with rate $\|\nabla f(\mathbf{p})\|$ and decreases most rapidly in the opposite direction with rate $-\|\nabla f(\mathbf{p})\|$.

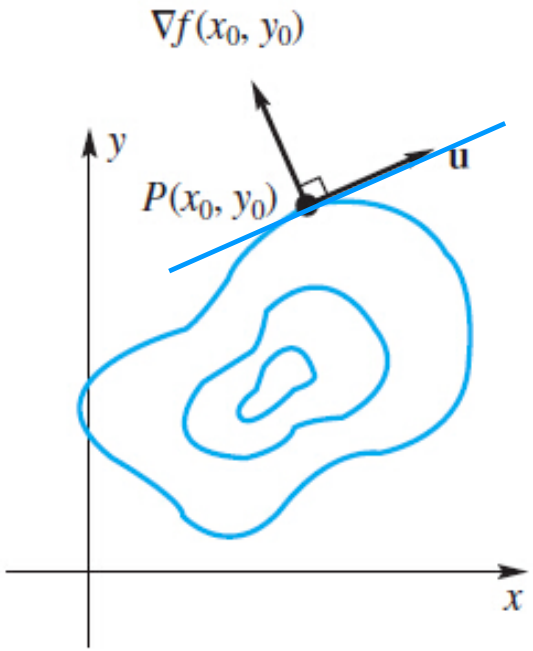


- **Level Curves and Gradients**

Thus ∇f and **u** are penpendicular.

Theorem C

The gradient of f at a point P is perpendicular to the level curve of f that goes through P. That is, the gradient ∇f is normal to level curves.



➢12.6 The Chain Rule(链式法则) and Implicit Differentiation

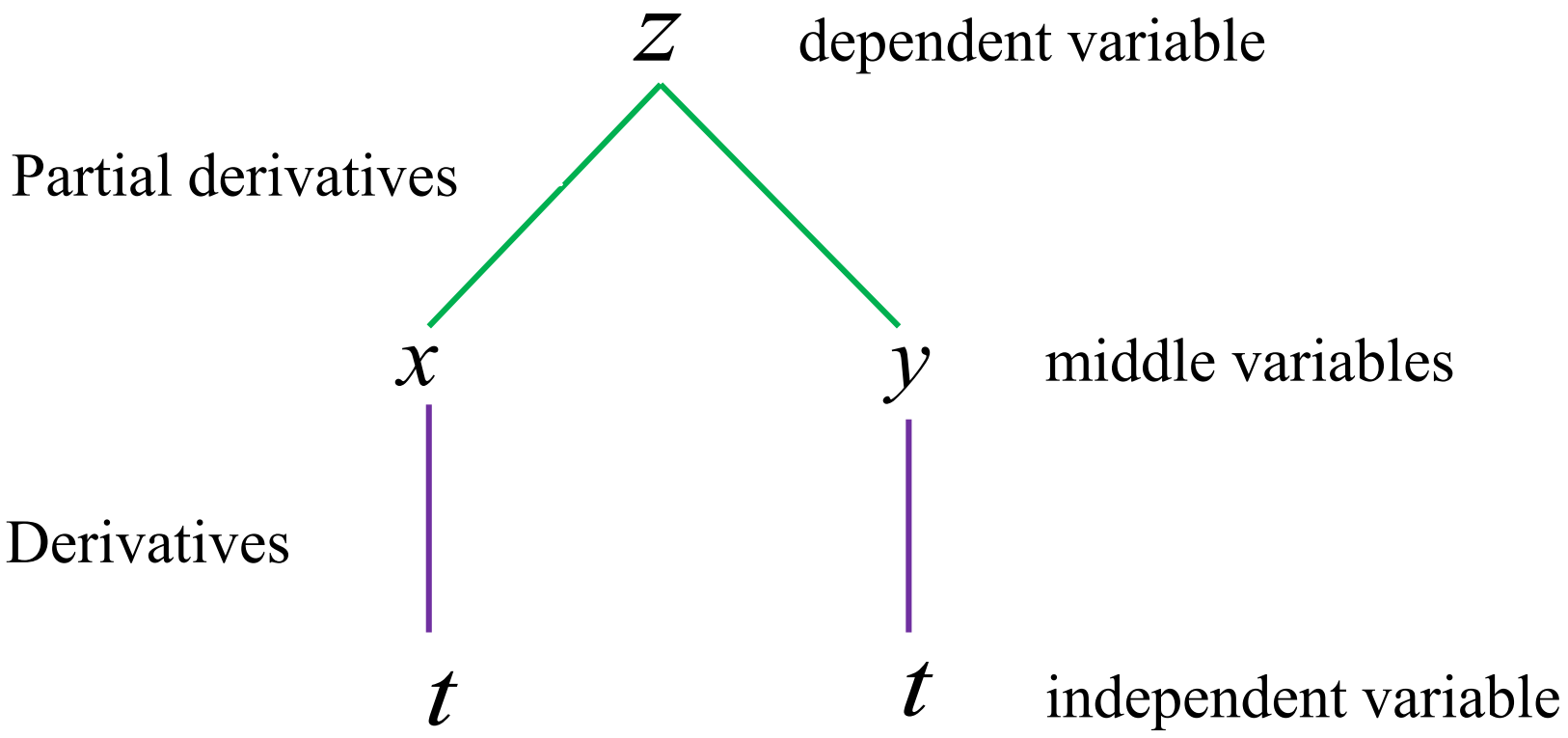
Theorem A **Chain Rule**

If $z = f(x, y)$ is a differentiable of x and y, where $x=g(t)$, $y=h(t)$ are both differentiable functions of t, then $z = f(g(t), h(t))$ is a differentiable function of t and

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.}$$

Tree Diagram

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.}$$



Second Version

If $z = f(x, y)$, $x = g(s, t)$, $y = h(s, t)$, then the composition

$z = f(g(s,t), h(s,t))$ is a function of two variables s, t.

For example $\quad z = e^x \sin y,\ x = st,\ y = s + t$

Composition $\quad z = e^{st} \sin(s + t)$

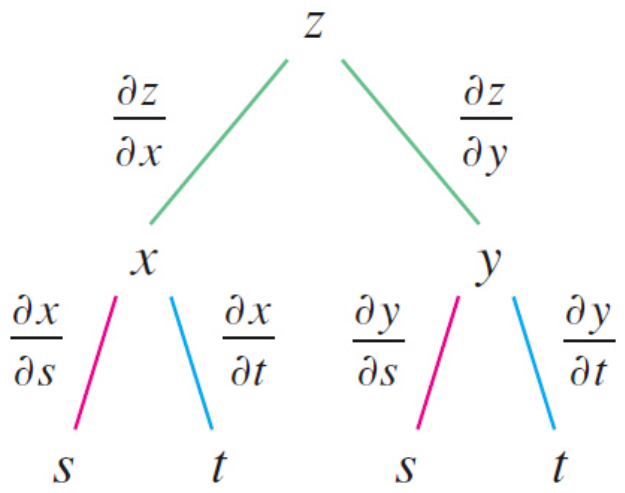
How to express $\dfrac{\partial z}{\partial s}, \dfrac{\partial z}{\partial t}$ in terms of

$$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial x}{\partial s}, \frac{\partial x}{\partial t}, \frac{\partial y}{\partial s}, \frac{\partial y}{\partial t}?$$

Theorem B Chain Rule

If $z = f(x, y)$ is a differentiable of x and y, where $x=g(s, t)$, $y=h(s, t)$ are both differentiable functions of s and t, then $z = f(g(s,t), h(s,t))$ is a differentiable function of s and t and

$$\boxed{\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}, \\ \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}. \end{aligned}}$$



- **Implicit Differentiation**

Case I Suppose that $F(x,y)=0$ defines y implicitly as a function of x, namely $y=f(x)$ where $F(x, f(x))=0$ for all x in the domain of f.

Since both x and y are functions of x, by applying chain rule to differentiate both side of the equation $F(x, y)=0$ we obtain

$$\frac{\partial F}{\partial x}\frac{dx}{dx}+\frac{\partial F}{\partial y}\frac{dy}{dx}=0.$$

If $\frac{\partial F}{\partial y}\neq 0$, we get

$$\boxed{\frac{dy}{dx}=-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}=-\frac{F_x}{F_y}.}$$

Case II Suppose that $F(x,y,z)=0$ defines z implicitly as a function of x and y, namely $z=f(x,y)$ where $F(x,y,f(x,y))=0$ for all (x,y) in the domain of f.

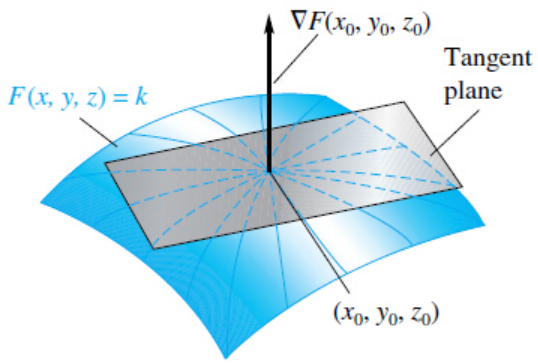
Differentiating both side of the equation $F(x, y, z)=0$ with respect to x by holding y fixed, we obtain

$$\frac{\partial F}{\partial x}\frac{\partial x}{\partial x}+\frac{\partial F}{\partial y}\frac{\partial y}{\partial x}+\frac{\partial F}{\partial z}\frac{\partial z}{\partial x}=0.$$

If $\frac{\partial F}{\partial z}\neq 0$, we get the first formula. The second formula is obtained in a similar manner.

$$\boxed{\frac{\partial z}{\partial x}=-\frac{F_x}{F_z},\quad \frac{\partial z}{\partial y}=-\frac{F_y}{F_z}.}$$

➢12.7 Tangent Planes and Differentials



The surface S: $F(x,y,z)=k$. $P(x_0,y_0,z_0)\in S$

Let C be any curve that lies in S and passes through P, which can be described by parametric equations $\mathbf{r}(t)=\langle x(t),y(t),z(t)\rangle\subset S$

Theorem A Equation of Tangent Plane（切平面）

$$F_x(x_0,y_0,z_0)(x-x_0)+F_y(x_0,y_0,z_0)(y-y_0)+F_z(x_0,y_0,z_0)(z-z_0)=0$$

Remarks

1. ∇F is a vector normal to the level surface of F.

2. For the surface $z=f(x,y)$, set $F(x,y,z)=f(x,y)-z$.

   Gradient of F is $\nabla F=\langle f_x, f_y, -1\rangle$.

   The equation of the tangent plane at $P(x_0,y_0,z_0)\in S$

$$f_x(x_0,y_0)(x-x_0)+f_y(x_0,y_0)(y-y_0)-(z-z_0)=0$$

3. Normal line(法线) of S through P is given by

$$\boxed{\frac{x-x_0}{F_x(x_0,y_0,z_0)}=\frac{y-y_0}{F_y(x_0,y_0,z_0)}=\frac{z-z_0}{F_z(x_0,y_0,z_0)}}$$

- **Differentials**

Definition If $z = f(x, y)$ and f is ***differentiable*** at (x_0, y_0), then

$$f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$$

is called the ***differential* of *f* at** (x_0, y_0). If f is a differentiable function, then (total) ***differential* (全变分) *dz*** of f is defined by

$$dz = f_x(x, y)dx + f_y(x, y)dy$$

where dx and dy are increments of independent variables.

- ***Differential*** of functions of three variables

If $u = f(x, y, z)$, then $du = f_x dx + f_y dy + f_z dz$.

- **Linear Approximation**

$$f(x,y) \approx f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) \quad (3)$$

Example Find an approximation of $(1.04)^{2.02}$.

Then an approximate formula is

$$f\left(x_0+\Delta x, y_0+\Delta y\right) \approx x_0^{\,y_0} + y_0 x_0^{\,y_0-1}\Delta x + x_0^{\,y_0} \ln x_0 \Delta y.$$

Putting $x_0 = 1, y_0 = 2, \Delta x = 0.04, \Delta y = 0.02.$

$$(1.04)^{2.02} \approx 1 + 2\cdot 0.04 + 0\cdot 0.02 = 1.08.$$

➢12.8 Maxima(最大值) and Minima(最小值)

1. **Local maxima and local minima**
2. **Global maxima and global minima**

Procedure for Finding Local Extrema of $z = f(x, y)$ on open set

Step 1 Solve the simultaneous equations $\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases}$

to identify all stationary points.

Step 2 Calculuate the second partial derivatives f_{xx}, f_{xy}, f_{yy} and $D(x, y)$ and evaluate the values of f_{xx} and $D(x, y)$ at each stationary point, then list a Table.

Step 3 Use the results in Table and Theorem C to determine whether f has local extreme value at each stationary point or not and evaluate local extreme values of f.

Sufficient Conditions for Extrema on open set

Theorem C Second Partials Test for Local Extreme Values

Suppose that $f(x, y)$ has continuous second partial derivatives in a neighborhood of (a, b) and that $\nabla f(a, b) = \mathbf{0}$. Let

$$D = D(a,b) = f_{xx}(a,b) f_{yy}(a,b) - (f_{xy}(a,b))^2$$

Then

(i) if $D > 0$ and $f_{xx}(a,b) < 0$, $f(a,b)$ is a **local maximum value**;

(ii) if $D > 0$ and $f_{xx}(a,b) > 0$, $f(a,b)$ is a **local minimum value**;

(iii) if $D < 0$, $f(a,b)$ is not an extreme value ((x_0, y_0) is a **saddle point**);

(iv) if $D = 0$, the test is **inconclusive**.

Note It is helpful to write D as a determinant (**Hession** of f)

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2.$$

Where Do Global Maximum and Minimum Value Occur?

Therem B Critical Point Theorem

Let f be defined on a set S containing $\mathbf{p}_0$. If $f(\mathbf{p}_0)$ is an extreme value, then $\mathbf{p}_0$ must be a critical point; that is, either $\mathbf{p}_0$ is

(i) a stationary point of f; or

(ii) a singular point of f; or

(iii) a boundary point of S.

Procedure for finding the maximum and minimum value on closed bounded set

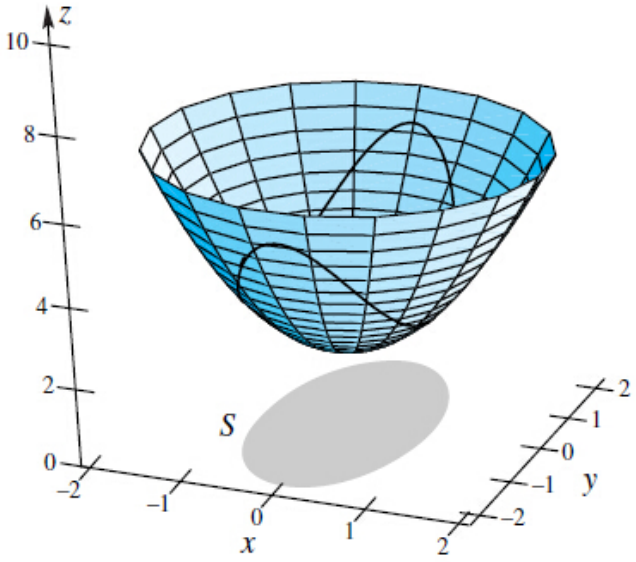
Step 1 Find the values of f at stationary points and singular points in S.

Step 2* Find the extreme values of f on the boundary of S.

Step 3 The largest of these values is the maximum value; the smallest is the minimum value.

Example 5 Find the maximum and minimum values of $f(x,y)=2+x^2+y^2$ on the closed and bounded set

$$S=\{(x,y): x^2+\frac{y^2}{4}\le 1\}.$$



Solution On the interior of S.

Find all stationary points.

$$\begin{cases} f_x = 2x = 0 \\ f_y = 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

There is only one stationary point (0,0) and no singular point.

$f(0,0)=2.$

On the boundary of S. Parametrize the boundary of S by

$$x=\cos t, y=2\sin t, \quad 0\le t\le 2\pi.$$

The optimization problem then reduces to one of optmizing the function of one variable

$$\begin{aligned} g(t) &= f(\cos t, 2\sin t) \\ &= 2 + \cos^2 t + 4\sin^2 t \\ &= 3 + 3\sin^2 t, \quad 0 \le t \le 2\pi. \end{aligned}$$

When $t = 0, \pi, 2\pi$, g has minimum value 3.

When $t = \pi/2, 3\pi/2$, g has maximum value 6.

The corresponding extreme values of f on the boundary are

$$\begin{aligned} f(1,0) &= 3 \qquad & f(0,2) &= 6 \\ f(-1,0) &= 3 \qquad & f(0,-2) &= 6 \end{aligned}$$

Therefore, the minimum value of f on S is 2, and the maximum value is 6.

12.9 The Method of Lagrange Multipliers

Question Find the extreme values of **objective function** $z = f(x, y)$ subject to the **constraint** $g(x, y) = 0$.

General Method The method of **Lagrange Multipliers**

Lagrange's Method (another form)

To find extreme values of $u = f(x, y)$ subject to the constraint $g(x, y) = 0$:

Step 1 Construct a Lagrange function

$L(x, y, \lambda) = f(x, y) - \lambda g(x, y)$

Step 2 Find all values of x, y, z and λ such that

$$\begin{cases} L_x = 0 : f_x(x, y) = \lambda g_x(x, y) \\ L_y = 0 : f_y(x, y) = \lambda g_y(x, y) \\ L_\lambda = 0 : g(x, y) = 0. \end{cases} \overset{(\text{step}1+2)}{\Leftrightarrow} \begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = 0 \end{cases}$$

Step 3 Evaluate f at all points (x, y) that result from **Step** 2.

Step 4 Compare the function values at these points to get the extreme values.

Theorem A Lagrange's Method (3-Dimensional case)

To find extreme values of $u = f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$:

Step 0 Construct a Lagrange function

$L(x, y, z, \lambda) = f(x, y, z) - \lambda g(x, y, z)$

Step 1 Find all values of x, y, z and λ such that

$$\begin{cases} L_x = 0 : f_x(x, y, z) = \lambda g_x(x, y, z) \\ L_y = 0 : f_y(x, y, z) = \lambda g_y(x, y, z) \\ L_z = 0 : f_z(x, y, z) = \lambda g_z(x, y, z) \\ L_\lambda = 0 : g(x, y, z) = 0. \end{cases} \overset{(\text{step } 0+1)}{\Leftrightarrow} \begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ g(x, y, z) = 0 \end{cases}$$

Step 2 Evaluate f at all points (x, y, z) that result from **Step** 1.

Step 3 Compare the function values at these points to get the extreme values.

The parameter λ is called the **Lagrange Multiplier**.

Optimizing a Function over a Closed and Bounded Set

Example 3 Find the maximum and minimum values of $f(x,y) = 4 + xy - x^2 - y^2$ on the set $S = \{(x,y) : x^2 + y^2 \le 1\}$.

Solution In the interior of S.

Find all stationary points.

$$\begin{cases} f_x = y - 2x = 0 \\ f_y = x - 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

There is only one stationary point (0,0) and no singular point.

$f(0,0) = 4$.

On the boundary of S:

The objective function $f(x,y) = 4 + xy - x^2 - y^2$.

Constraint: $g(x,y) = x^2 + y^2 - 1 = 0$.

$$\begin{cases} \nabla f = \lambda \nabla g \Rightarrow \langle y-2x, x-2y \rangle = \lambda \langle 2x, 2y \rangle \\ g(x,y,z) = 0 \end{cases} \Rightarrow \begin{cases} y - 2x = \lambda 2x & (1) \\ x - 2y = \lambda 2y & (2) \\ x^2 + y^2 - 1 = 0. & (3) \end{cases}$$

$$\Rightarrow \frac{y-2x}{x-2y} = \frac{x}{y} (y \neq 0) \Rightarrow x^2 = y^2 \Rightarrow x = \pm y, x = \frac{\sqrt{2}}{2} \text{ or } -\frac{\sqrt{2}}{2}.$$

Four critical points:

$$(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}),\ (\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}),\ (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}),\ (-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}).$$

$$f(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}) = f(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}) = \frac{7}{2}; f(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}) = f(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}) = \frac{5}{2}.$$

Since the maximum and minimum both exist, we conclude that the minimum value of f on S is $5/2$, and the maximum value on S is 4.